

# Supplementary Material: FactorJEPA: World Models That Predict Crowded, Chaotic City Streets

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## Appendix

This appendix provides the complete evidence and implementation record underlying the main paper. Its organization follows the scientific progression of the study. Appendices A–C establish the provenance, distinctiveness, and reproducibility of DENSEWORLD, its factor targets, and the executed training protocols. Appendices D and E then examine the central methodological claims: which components produce the observed gains, and whether the evaluation is statistically reliable and robust.

The appendix distinguishes primary confirmatory evidence from diagnostic and robustness analyses. Unless otherwise stated, comparisons use identical evaluation clips, paired resampling units, and the same independently defined audit signals. Factor separation is interpreted operationally at the level of predictive channels rather than as coordinate-level identifiability or complete statistical independence. Causal future-block evaluation measures predictive accuracy on the future temporal half, while cross-scale replication refers specifically to stability between the 1B and 2B V-JEPA 2.1 backbones.

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training, validation, and teacher-relative test targets. No manually annotated semantic labels are used.

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## A DENSEWORLD Construction, Splits, and Responsible Use

DENSEWORLD comprises approximately **276 hours** of drive-through, walk-through, and aerial video acquired across **22 Indian cities**. It targets predictive modeling under high agent density, heterogeneous traffic, persistent occlusion, soft spatial boundaries, and frequent local interaction. DENSEWORLD contains *no manually annotated semantic labels*; all layout, agent, visibility, and interaction targets are produced automatically using a fixed Grounding DINO + SAM 3 pipeline.

### A.1 Acquisition Protocol and Capture Modes

DENSEWORLD was assembled from **714 long-form YouTube videos** (480p) selected under a **shared coverage specification**. The specification defined coverage targets over geographic sites, scene types, times of day, weather, visibility, road structure, and infrastructure conditions. It controlled *where and how* videos were acquired without scripting traffic, pedestrian, or animal behavior.

The collection uses three capture modes. **Drive-through** recordings emphasize road-level ego-motion, mixed traffic, and near-field interactions. **Walk-through** recordings capture pedestrian-scale navigation, shared spaces, markets, commercial corridors, and transit areas. **Aerial** recordings provide broader context for crowd flow, junction organization, traffic structure, and interaction topology. City, collection session, capture mode, and source timestamps are retained as acquisition metadata.

### A.2 Source Processing, Clip Construction, and Provenance

Source videos are decoded using FFmpeg<sup>1</sup> while retaining frame timestamps and source identifiers. Shot boundaries are detected using the PySceneDetect AdaptiveDetector<sup>2</sup>, which normalizes adjacent-frame changes in hue, saturation, and luminance by a rolling temporal average. This reduces false boundaries caused by rapid ego-motion and camera shake.

We use a fixed corpus-wide configuration: `adaptive_threshold= 3.0`, `min_content_val= 15.0`, `window_width= 2`, and `min_scene_len= 15` frames. For a source video  $v = \{x_t^{(v)}\}_{t=1}^{T_v}$ , the detector returns ordered boundaries

$$\mathcal{B}_v = \left\{ 1 = b_0^{(v)} < b_1^{(v)} < \dots < b_{J_v}^{(v)} = T_v \right\},$$

which partition the source into candidate visually continuous shots. Shots shorter than the minimum usable duration are excluded; retained shots yield model-ready clips of approximately 6–13 seconds. No clip crosses a detected shot boundary.

The retained provenance hierarchy is

city  $\rightarrow$  collection session  $\rightarrow$  source video  $\rightarrow$  shot  $\rightarrow$  clip

<sup>1</sup><https://ffmpeg.org/>

<sup>2</sup><https://www.scenesdetect.com/docs/latest/api/detectors.html>

For every clip  $c$ , we store

$$\pi(c) = (i_{\text{city}}, i_{\text{session}}, i_{\text{source}}, i_{\text{shot}}, m_{\text{capture}}, t_{\text{start}}, t_{\text{end}}).$$

Thus, the clip is the **computational input**, while its source, session, and city identifiers preserve higher-level geographic and temporal dependence.

### A.3 Partitioning and Factor Target Provenance

Source videos are assigned as indivisible groups using an **source-grouped train/validation/test allocation** (86,831 / 909 / 23,106 clips; 4,841 clips without factor targets excluded), stratified by city and capture mode. Split assignment precedes clip extraction, so every shot and clip derived from one source video inherits the same partition. Let  $\mathcal{V}_s$  denote the source-video set assigned to partition  $s$ . The executed grouping satisfies

$$\mathcal{V}_{\text{train}} \cap \mathcal{V}_{\text{val}} = \mathcal{V}_{\text{train}} \cap \mathcal{V}_{\text{test}} = \mathcal{V}_{\text{val}} \cap \mathcal{V}_{\text{test}} = \emptyset.$$

DENSEWORLD does not use manually annotated factor labels. For each partition, automated structural targets are generated by the fixed Grounding DINO + SAM 3 pipeline:

$$\mathcal{T}_s^{\text{DINO}} = \{ \Psi_{\text{struct}}(F_{\text{factor}}(x)) : x \in \mathcal{D}_s \}, \\ s \in \{\text{train, val, test}\}.$$

Training targets may provide factor supervision; validation targets may support prespecified selection and calibration; and test targets are used only for held-out *teacher-relative* factor diagnostics. Test-derived targets, thresholds, and metrics never influence gradient updates or model selection.

### A.4 Corpus Composition and Coverage

**Geographic and capture-mode composition.** The collection spans metropolitan and additional urban sites across India. These groups are **collection strata**, not claims of an official administrative taxonomy or within-stratum homogeneity. Figure 1 reports the contribution of drive-through, walk-through, and aerial clips for each city. The city-by-mode distribution is observational and intentionally heterogeneous; bar length describes corpus composition, not evaluation weight or statistical independence.

**Environmental and agent coverage.** DENSEWORLD includes commercial, residential, transit, heritage, coastal, and high-density street environments, spanning markets, junctions, flyovers, highways, promenades, bazaars, ghats, beaches, and shared roads. It also captures variation in illumination, weather, traffic composition, crowd density, road geometry, surface quality, encroachment, and pedestrian-vehicle separation.

The dynamic-agent taxonomy extends beyond passenger vehicles and pedestrians to include auto-rickshaws, cycle rickshaws, delivery riders, multi-rider two-wheelers, cyclists, mobile vendors, animal-drawn vehicles, and independently moving animals. Figure 2 illustrates category semantics; it is neither a frequency-proportional sample nor manually annotated evidence. Quantitative prevalence and heterogeneity statistics derived from the automated pipeline are reported as *automatically estimated* quantities.

Table 1: **Composition, information flow, and independence constraints of the DENSEWORLD partitions.** Allocation is performed over source-video groups. Durations are approximate because grouped recordings vary in length and usable-shot yield. Grounding DINO + SAM 3 test measurements are reported as teacher-relative; headline prediction, motion, and RGB metrics use fixed evaluators that do not provide FactorJEPA’s factor-supervision targets.

| Partition    | Share       | Duration         | Gradient updates | Model selection | Threshold calibration | Signal exposed                                                                                                      | Isolation constraint                                                                                  |
|--------------|-------------|------------------|------------------|-----------------|-----------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Training     | 80%         | ≈ 800 h          | Yes              | No              | No                    | JEPA prediction targets and automatically derived layout, agent, and interaction targets.                           | Source-video disjoint from validation and test; all descendants of a source remain grouped.           |
| Validation   | 10%         | ≈ 100 h          | No               | Yes             | Yes                   | Prespecified validation metrics and automatically derived targets used only for selection and calibration.          | Source-video disjoint from training and test; never used for gradient optimization.                   |
| Test         | 10%         | ≈ 100 h          | No               | No              | No                    | Frozen headline evaluators and automatically derived targets used only for explicitly teacher-relative diagnostics. | Source-video disjoint from training and validation; all configurations are frozen before test access. |
| <b>Total</b> | <b>100%</b> | <b>≈ 1,000 h</b> | <b>–</b>         | <b>–</b>        | <b>–</b>              | <b>–</b>                                                                                                            | <b>–</b>                                                                                              |

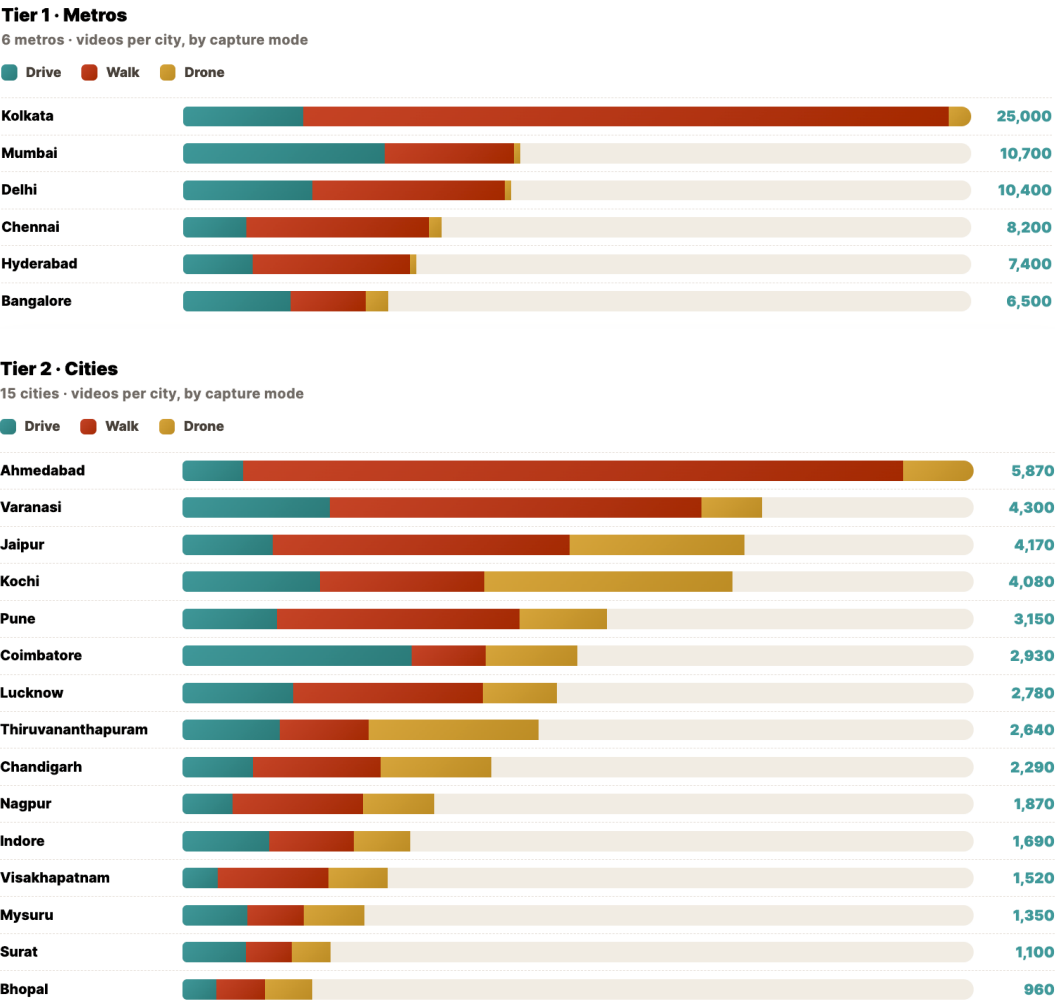


Figure 1: **City-level composition by capture mode.** The upper panel shows metropolitan collection sites and the lower panel shows additional urban sites. Each bar decomposes a city’s model-ready clips into drive-through, walk-through, and aerial capture. The value at the right gives the city total; both panels use a common horizontal scale.



Figure 2: **Illustrative DENSEWORLD dynamic-agent taxonomy.** The examples communicate the semantic scope of the automated taxonomy, including conventional transport, intermediate mobility, mobile vendors, animal-drawn transport, and independently moving animals. They are illustrative category renderings rather than frequency-proportional dataset samples.

### A.5 Privacy Processing, Responsible Use, and Research Artifacts

**Privacy processing.** Before downstream use, clips pass through an automated privacy pipeline. Faces are localized with SCRFD<sup>3</sup>, and vehicle-registration plates are localized with a custom-trained YOLO detector<sup>4</sup>. Detections are associated temporally using ByteTrack<sup>5</sup>, expanded by a fixed spatial safety margin, and redacted using OpenCV GaussianBlur<sup>6</sup>.

The policy covers frontal, profile, small, and partially occluded faces, detectable faces in reflections, and readable or partially readable vehicle-registration plates. Let  $\mathcal{R}_t^{\text{face}}$  and  $\mathcal{R}_t^{\text{plate}}$  denote the temporally associated and expanded regions in frame  $x_t$ . The processed frame is

$$\tilde{x}_t = \mathcal{B}_\sigma \left( x_t, \mathcal{R}_t^{\text{face}} \cup \mathcal{R}_t^{\text{plate}} \right),$$

<sup>3</sup><https://github.com/deepinsight/insightface/tree/master/detection/scrfd>

<sup>4</sup><https://docs.ultralytics.com/tasks/detect/>

<sup>5</sup><https://github.com/FoundationVision/ByteTrack>

<sup>6</sup>[https://docs.opencv.org/4.x/d4/d86/group\\_\\_imgproc\\_\\_filter.html](https://docs.opencv.org/4.x/d4/d86/group__imgproc__filter.html)

where  $\mathcal{B}_\sigma$  applies region-scaled Gaussian filtering. Automated quality control checks track discontinuities, unmatched detections, frame-boundary truncation, and processing failures. Segments with incomplete or failed redaction are excluded. Only  $\tilde{x}_t$  enters target construction, training, evaluation, or external illustration.

**Responsible use and research artifacts.** DENSEWORLD is intended for predictive world modeling, video representation learning, multi-agent forecasting, and robustness under partial observability. It is not intended for person identification, persistent individual tracking, sensitive-attribute inference, or surveillance.

The complete corpus is maintained within the **governed research environment** used for the reported experiments. Supporting artifacts include the FactorJEPa implementation, preprocessing configuration, factor target-construction pipeline, metric definitions, automated taxonomy, aggregate corpus statistics, and representative privacy-filtered material, subject to applicable rights and privacy review.

## B Five-Axis Validation of the DENSEWORLD Regime

We ask whether DENSEWORLD remains quantitatively distinct from BDD100K (Yu et al. 2020) and nuScenes (Caesar et al. 2020) after controlling for observable differences in capture and scene composition. We characterize this regime along five axes: **agent count density**, **agent occupancy**, **visibility pressure**, **interaction pressure**, and **agent heterogeneity**.

The comparison is **outcome-blind**, **scene-matched**, and **dependence-aware**. All datasets are processed using the same frozen Grounding DINO + SAM 3 measurement pipeline; none of the five axes enters the matching procedure; and uncertainty is clustered above the frame and clip levels. The resulting quantities are therefore interpreted as *teacher-relative dataset statistics*, not manually annotated ground truth or causal effects of dataset membership.

### B.1 Shared Measurement Protocol and Automated Taxonomy

All three datasets are processed through the common sequence

video  $\rightarrow$  sampled frames  $\rightarrow$  Grounding DINO + SAM 3 detections  $\rightarrow$  structural pseudo-annotations  $\rightarrow$  five-axis statistics

For frame  $x_t^{(d)}$  from dataset  $d \in \{\text{DW}, \text{BDD}, \text{NS}\}$ , the frozen Grounding DINO + SAM 3 pipeline (Carion et al. 2025) and structural postprocessor produce

$$\begin{aligned} \mathcal{P}_t^{(d)} &= \Psi_{\text{struct}} \left( F_{\text{factor}}(x_t^{(d)}) \right) \\ &= \left\{ \left( \hat{c}_{i,t}, \hat{s}_{i,t}, \hat{B}_{i,t}, \hat{M}_{i,t}, \hat{\ell}_{i,t} \right) \right\}_{i \in \mathcal{I}_t^{(d)}}. \end{aligned}$$

Here,  $\mathcal{I}_t^{(d)}$  indexes candidate instances;  $\hat{c}_{i,t}$ ,  $\hat{s}_{i,t}$ ,  $\hat{B}_{i,t}$ , and  $\hat{M}_{i,t}$  denote the predicted class, confidence, bounding box, and visible instance mask; and  $\hat{\ell}_{i,t}$  is the temporally associated track identifier.

SAM 3 supplies text-prompted instance masks directly, and  $\Psi_{\text{struct}}$  denotes the executed frozen sequence of proposal extraction, class assignment, mask construction, duplicate suppression, and temporal association. Its component architectures, checkpoint identifiers, inference resolution, thresholds, suppression rule, and association parameters are fixed before comparison and applied identically to all datasets.

Let  $\Omega_t$  denote the valid, unpadded image region. The retained agent set is

$$\mathcal{A}_t^{(d)} = \left\{ i \in \mathcal{I}_t^{(d)} : \hat{s}_{i,t} \geq \tau_{\text{det}}, \frac{|\hat{M}_{i,t} \cap \Omega_t|}{|\Omega_t|} \geq a_{\min} \right\}.$$

The same temporal stride, resizing rule, confidence threshold, minimum-area criterion, and temporal-association configuration are used for DENSEWORLD, BDD100K, and nuScenes. Predictions that cannot be mapped unambiguously are excluded rather than reassigned using dataset-specific heuristics.

Predicted classes are mapped to seven shared superclasses:

- (i) **pedestrian**
- (ii) **pedal cycle**
- (iii) **powered two-wheeler**
- (iv) **light passenger vehicle**
- (v) **heavy or public vehicle**
- (vi) **intermediate or non-motorized transport**
- (vii) **animal or animal-drawn transport**

Native BDD100K and nuScenes annotations are not used to compute, calibrate, or select any primary statistic. To expose possible teacher-induced domain effects, we report the retained-instance rate, unresolved-class rate, confidence distribution, small-instance exclusion rate, and track-fragmentation rate separately for each dataset. The primary analysis is additionally repeated over a prespecified grid of confidence, area, and association settings.

### B.2 Definition of the Five Regime Axes

Suppressing the dataset superscript for clarity, let

$$N_t = |\mathcal{A}_t|, \quad \mathcal{T}_1 = \{t : N_t \geq 1\}, \quad \mathcal{T}_2 = \{t : N_t \geq 2\}.$$

All boxes and masks are clipped to  $\Omega_t$  before measurement. Count and occupancy are evaluated over all sampled frames. Visibility pressure is evaluated over  $\mathcal{T}_1$ , whereas interaction and heterogeneity are evaluated over  $\mathcal{T}_2$ . We report the eligible-frame rate for every conditional statistic.

#### 1. Agent count density.

The primary statistic is the mean number of retained dynamic agents per sampled frame:

$$D_{\text{count}} = \frac{1}{T} \sum_{t=1}^T N_t.$$

We additionally report valid-support-normalized count,

$$D_{\text{count}}^{\text{area}} = \frac{1}{T} \sum_{t=1}^T \frac{N_t}{|\Omega_t|/10^6},$$

as a sensitivity measure for cropping, padding, and inference resolution. It is not interpreted as physical agent density.

#### 2. Agent occupancy.

Occupancy is the fraction of valid image support covered by the union of visible agent masks:

$$D_{\text{occ}} = \frac{1}{T} \sum_{t=1}^T \frac{\left| \bigcup_{i \in \mathcal{A}_t} (\hat{M}_{i,t} \cap \Omega_t) \right|}{|\Omega_t|}.$$

Taking the union prevents crowded or overlapping regions from being counted multiple times.

#### 3. Visibility pressure.

Because no amodal annotations are available, this axis is defined as an automated *visibility-pressure proxy*. It combines three observable cues: substantial pairwise box

overlap, valid-image-boundary truncation, and short interior track disappearance followed by reappearance. For boxes  $B_i$  and  $B_j$ , define intersection over minimum area as

$$\text{IoM}(B_i, B_j) = \frac{|B_i \cap B_j|}{\min(|B_i|, |B_j|) + \epsilon}.$$

The agent-level visibility indicator is

$$z_{i,t} = \mathbf{1} \left[ \max_{j \neq i} \text{IoM}(\hat{B}_{i,t}, \hat{B}_{j,t}) \geq \tau_{\text{ov}} \vee e_{i,t} = 1 \vee g_{i,t} = 1 \right]$$

where  $e_{i,t}$  indicates boundary truncation and  $g_{i,t}$  indicates an interior track gap followed by reappearance within the prespecified temporal window. Visibility pressure is

$$D_{\text{vis}} = \frac{1}{|\mathcal{T}_1|} \sum_{t \in \mathcal{T}_1} \frac{\sum_{i \in \mathcal{A}_t} z_{i,t}}{N_t}.$$

Overlap, boundary-truncation, and track-gap rates are also examined separately to verify that the composite is not dominated by one proxy.

#### 4. Interaction pressure.

We construct a frame-level interaction graph  $\mathcal{G}_t = (\mathcal{A}_t, \mathcal{E}_t)$ . Let  $\mathbf{p}_{i,t}$  denote the normalized image-plane center of agent  $i$ . Velocities  $\tilde{\mathbf{u}}_{i,t}$  are computed after removing estimated global image motion. Define

$$\mathbf{r}_{ij,t} = \mathbf{p}_{j,t} - \mathbf{p}_{i,t}, \quad \mathbf{w}_{ij,t} = \tilde{\mathbf{u}}_{j,t} - \tilde{\mathbf{u}}_{i,t}.$$

For approaching pairs, the predicted time and distance at closest approach are

$$t_{ij,t}^* = -\frac{\mathbf{r}_{ij,t}^\top \mathbf{w}_{ij,t}}{\|\mathbf{w}_{ij,t}\|_2^2 + \epsilon},$$

$$d_{ij,t}^* = \|\mathbf{r}_{ij,t} + t_{ij,t}^* \mathbf{w}_{ij,t}\|_2.$$

An interaction edge is present when the pair is currently proximate or is predicted to approach within both the temporal and spatial thresholds:

$$(i, j) \in \mathcal{E}_t \iff \|\mathbf{r}_{ij,t}\|_2 \leq \rho \vee (0 < t_{ij,t}^* \leq \tau_t \wedge d_{ij,t}^* \leq \tau_d).$$

Interaction pressure is the conditional mean graph degree:

$$D_{\text{int}} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} \frac{2|\mathcal{E}_t|}{N_t}.$$

Per-pair edge density is retained as a sensitivity statistic to separate interaction structure from the mechanical effect of agent count.

#### 5. Agent heterogeneity.

Let  $N_{c,t}$  be the number of retained agents assigned to superclass  $c \in \mathcal{C}$ . The primary statistic is the probability that two agents drawn without replacement from the same frame belong to different superclasses:

$$h_t = 1 - \sum_{c \in \mathcal{C}} \frac{N_{c,t}(N_{c,t} - 1)}{N_t(N_t - 1)}, \quad t \in \mathcal{T}_2.$$

Dataset-level heterogeneity is

$$D_{\text{het}} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} h_t.$$

This pairwise definition lies in  $[0, 1]$  and is less sensitive to small per-frame counts than plug-in entropy. Normalized entropy, class richness, and the effective number of classes are retained as secondary measures.

### B.3 Matched Cross-Dataset Comparison

The primary analysis matches DENSEWORLD drive-through clips separately against road-level clips from BDD100K (Yu et al. 2020) and nuScenes (Caesar et al. 2020). Walk-through and aerial clips are summarized separately and do not enter the matched driving-benchmark comparison.

Matching is performed without access to the five axes or their component measurements. Clips are first assigned to common strata defined by capture mode, scene family, illumination, weather, and road context. These nuisance attributes are obtained using the same fixed metadata rules across datasets. Within each stratum, we perform 1:1 nearest-neighbor matching without replacement over standardized clip duration, valid image support, aspect ratio, resolution, and common camera metadata. Covariates unavailable consistently across all datasets are excluded from the primary matching model.

Let  $\mathbf{z}_v$  denote the matching covariates for clip  $v$ . Post-matching balance is measured using the standardized mean difference

$$\text{SMD}(z) = \frac{\bar{z}_{\text{DW}} - \bar{z}_{\text{ref}}}{\sqrt{(s_{\text{DW}}^2 + s_{\text{ref}}^2)/2}}.$$

The primary analysis retains common-support strata satisfying

$$\max_{z \in \mathbf{z}} |\text{SMD}(z)| < 0.1.$$

We report the number of candidate and retained clips, independent source groups, unmatched fraction, and maximum absolute SMD before and after matching. Unmatched clips are excluded from the primary cross-dataset estimate and retained only in DENSEWORLD-specific coverage summaries.

For axis  $k$ , let  $\mathcal{T}_{k,u}$  be the eligible sampled frames from matched clips belonging to source group  $u$ . The source-level summary is

$$\bar{m}_{k,u}^{(d)} = \frac{1}{|\mathcal{T}_{k,u}|} \sum_{t \in \mathcal{T}_{k,u}} m_{k,t}^{(d)}.$$

The matched-population estimate gives equal weight to every retained source group:



Table 2: **Scene-matched agent density and occupancy.** Count density is measured in agents per sampled frame; occupancy is the percentage of valid image area covered by the union of visible agent masks.  $\Delta = \text{DW} - \text{reference}$ , and  $R$  is the ratio of the two means. DENSEWORLD results are aggregated over  $K$  executed folds. The final row gives the unweighted mean across the six matched scene strata.

| Matched scene stratum     | Agent count density (agents/frame) |                     |              |              | Agent occupancy (% valid area) |                     |               |              |
|---------------------------|------------------------------------|---------------------|--------------|--------------|--------------------------------|---------------------|---------------|--------------|
|                           | Matched reference                  | DENSEWORLD [K-fold] | $\Delta$     | $R$          | Matched reference              | DENSEWORLD [K-fold] | $\Delta_{pp}$ | $R$          |
| Residential lane          | 1.8                                | <b>3.3</b>          | +1.5         | 1.83×        | 2.0%                           | <b>3.5%</b>         | +1.5          | 1.75×        |
| Promenade                 | 1.4                                | <b>2.5</b>          | +1.1         | 1.79×        | 1.5%                           | <b>2.8%</b>         | +1.3          | 1.87×        |
| Market                    | 4.6                                | <b>12.4</b>         | +7.8         | 2.70×        | 4.2%                           | <b>11.1%</b>        | +6.9          | 2.64×        |
| Heritage / tourist        | 1.7                                | <b>3.0</b>          | +1.3         | 1.76×        | 0.9%                           | <b>1.4%</b>         | +0.5          | 1.56×        |
| Flyover / underpass       | 2.6                                | <b>5.0</b>          | +2.4         | 1.92×        | 8.8%                           | <b>20.8%</b>        | +12.0         | 2.36×        |
| Commercial                | 4.1                                | <b>11.0</b>         | +6.9         | 2.68×        | 3.9%                           | <b>9.0%</b>         | +5.1          | 2.31×        |
| <b>Equal-stratum mean</b> | <b>2.70</b>                        | <b>6.20</b>         | <b>+3.50</b> | <b>2.30×</b> | <b>3.55%</b>                   | <b>8.10%</b>        | <b>+4.55</b>  | <b>2.28×</b> |

*Interpretation.* The ratios in the final row are ratios of the equal-stratum means, not means of the six stratum-specific ratios. These quantities characterize the measured operating regime; larger values do not denote conventional benchmark performance.

$$\hat{\mu}_{r,k}^{(d)} = \frac{1}{|\mathcal{U}_{d,r}|} \sum_{u \in \mathcal{U}_{d,r}} \bar{m}_{k,u}^{(d)},$$

where  $r \in \{\text{BDD100K}, \text{nuScenes}\}$  identifies the reference-specific matched population. The DENSEWORLD estimate is therefore recomputed for its BDD100K-matched and nuScenes-matched subsets.

For each reference  $r$ , we report the absolute difference and ratio

$$\Delta_{r,k} = \hat{\mu}_{r,k}^{(\text{DW})} - \hat{\mu}_{r,k}^{(r)}, \quad R_{r,k} = \frac{\hat{\mu}_{r,k}^{(\text{DW})} + \epsilon}{\hat{\mu}_{r,k}^{(r)} + \epsilon}.$$

These contrasts describe separation under matched observable conditions; they are not interpreted causally.

#### B.4 Comparative Results, Uncertainty, and Sensitivity

The primary five-axis comparison is reported below. Entries follow the form  $X [X, X]$ : a fold-aggregated mean followed by its hierarchical 95% confidence interval. The bracketed value after **DENSEWORLD** denotes the number of executed folds.

Confidence intervals are estimated using a **paired bootstrap** that resamples the compared clips with replacement and applies the same resampled clips to all datasets in a contrast, re-estimating matching inside every replicate. We report 95% bias-corrected and accelerated (BCa) intervals for dataset means, absolute differences, and ratios.

Robustness is evaluated over prespecified variations of  $\tau_{\text{det}}$ ,  $a_{\text{min}}$ ,  $\tau_{\text{ov}}$ ,  $\rho$ ,  $(\tau_t, \tau_d)$ , and temporal sampling rate. We also compare source and equal-stratum weighting, conditional and unconditional normalization, and alternative common-support calipers.

Two automated negative controls test metric specificity. *Spatial permutation* preserves frame-level counts and class

frequencies while disrupting proximity and closest-approach edges. *Temporal permutation* preserves frame-level count and occupancy while disrupting motion consistency and track-gap structure. Each metric must attenuate under its corresponding control without mechanically altering the axes that the control is intended to preserve.

An axis is interpreted as exhibiting stable regime separation only when its effect direction is preserved across the pre-specified sensitivity grid and its bootstrap interval excludes zero. The resulting evidence is specific to the shared automated measurement system and matched observable conditions. It does not imply exhaustive geographic coverage, causal effects of dataset membership, or universal superiority over existing driving benchmarks.

## C Segmentation and Training

This appendix specifies the model-side supervision and executed training protocol for FactorJEPA. DENSEWORLD contains no manually annotated factor labels: layout, agent, visibility, and interaction targets are constructed automatically using a fixed Grounding DINO + SAM 3 teacher pipeline. These targets provide structured supervision during training and teacher-relative diagnostics during evaluation; they are distinct from the frozen evaluators used for headline future-prediction, motion, and RGB metrics.

We first document target construction and reliability weighting, then specify the factor-target construction, controlled baselines, and resource accounting. Table 4 defines the complete target contract; the executed training procedure is described in the main text.

### C.1 Adaptation Baselines: Formal Definitions

We compare FactorJEPA against three protocol-matched adaptation baselines. All retain the executed V-JEPA objective and train on the same raw clips; only the parameter-update mechanism changes (Table 3).



Table 3: **Protocol-matched V-JEPA adaptation baselines.** All methods retain the JEPA objective and train on the same raw clips; only the adaptation mechanism changes.

| Method                            | Adaptation mechanism                                                                                 | Question tested                                                 |
|-----------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>LoRA</b> (Hu et al. 2022)      | Adds low-rank updates, $\Delta W = (\alpha/r)BA$ , to selected frozen projections.                   | Is parameter-efficient low-rank adaptation sufficient?          |
| <b>DoRA</b> (Liu et al. 2024)     | Separates weight magnitude from direction and applies low-rank updates to the directional component. | Does weight decomposition recover structure missed by LoRA?     |
| <b>Auto-RGN</b> (Lee et al. 2023) | Selects transformer blocks using their relative gradient norms and updates only the selected subset. | Can gradient-guided selection localize the required adaptation? |

For LoRA, an adapted weight matrix is

$$W' = W_0 + \frac{\alpha}{r}BA, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, \quad A \in \mathbb{R}^{r \times d_{\text{in}}},$$

where  $r \ll \min(d_{\text{in}}, d_{\text{out}})$ . DoRA further separates the magnitude and direction of each weight vector:

$$W' = m \odot \frac{W_0 + \Delta W}{\|W_0 + \Delta W\|}, \quad \Delta W = \frac{\alpha}{r}BA,$$

allowing directional adaptation without coupling it to weight magnitude. Following the relative-gradient-norm criterion of Lee et al. (2023), our blockwise Auto-RGN implementation scores transformer block  $\ell$  as

$$s_\ell = \frac{\|\nabla_{\theta_\ell} \mathcal{L}_{\text{JEPA}}\|_2}{\|\theta_\ell\|_2 + \epsilon}, \quad \mathcal{S}_K = \text{TopK}_\ell(s_\ell),$$

and updates only  $\{\theta_\ell : \ell \in \mathcal{S}_K\}$ . Normalization by parameter magnitude prevents larger blocks from being favored solely because of scale. LoRA and DoRA target the same projection families, while Auto-RGN receives a matched trainable-parameter budget.

## C.2 Frozen Teacher and Factor-Target Construction

**Region extraction.** For each privacy-filtered clip  $\tilde{x}_n$ , the frozen Grounding DINO + SAM 3 pipeline (Carion et al. 2025) produces spatiotemporal visual features. The executed preprocessing operator converts these features into region-level pseudo-annotations:

$$\mathcal{P}_n = \mathcal{D}_{\text{pre}}(F_{\text{factor}}(\tilde{x}_n)) = \left\{ \left( m_{n,t}^{(i)}, b_{n,t}^{(i)}, u_{n,t}^{(i)}, p_{n,t}^{(i)} \right) \right\}_{t,i}.$$

Here,  $m_{n,t}^{(i)}$ ,  $b_{n,t}^{(i)}$ ,  $u_{n,t}^{(i)}$ , and  $p_{n,t}^{(i)}$  denote the visible mask, bounding box, visual descriptor, and confidence of region  $i$  at time  $t$ . The Grounding DINO + SAM 3 checkpoint, extracted feature layers, inference resolution, spatial stride, normalization, region-construction rule, and confidence filtering are fixed before model training and shared by all splits.

**Temporal association.** The association operator  $\mathcal{A}$  links compatible regions into tracklets:

$$\mathcal{A}(\mathcal{P}_n) = \left\{ \tau_n^{(i)} \right\}_{i=1}^{N_n^A}, \quad \tau_n^{(i)} = \left\{ m_{n,t}^{(i)}, b_{n,t}^{(i)}, u_{n,t}^{(i)} \right\}_{t=t_s^{(i)}}^{t_e^{(i)}}.$$

Association combines region appearance, mask or box overlap, and motion continuity. Track initiation, termination, admissible temporal gaps, and re-identification thresholds are held fixed across training, validation, and test. Short gaps may preserve a track identity; gaps beyond the admissible window terminate the tracklet.

**Structural targets.** A deterministic structural map converts region geometry, temporal continuity, visibility evidence, and relative motion into four targets:

$$\Psi_{\text{struct}}(\mathcal{P}_n, \mathcal{A}(\mathcal{P}_n)) \mapsto \{(T_{n,k}, q_{n,k})\}_{k \in \{L, A, V, I\}}.$$

The targets  $T_{n,L}$ ,  $T_{n,A}$ ,  $T_{n,V}$ , and  $T_{n,I}$  encode layout, agents, visibility, and tracklet-derived interactions. Each  $q_{n,k} \in [0, 1]$  is a reliability weight constructed from the underlying region confidence and temporal consistency. Low-confidence targets are down-weighted rather than converted into negative labels; targets below the exclusion threshold do not contribute to the corresponding factor loss.

**Target coverage and sensitivity.** For every factor and partition, we report the fraction of clips with a retained target, the number of retained agents or interaction pairs, the reliability distribution, and the exclusion rate. Threshold sensitivity varies the region-confidence, minimum-area, temporal-consistency, and pair-selection criteria while holding the model and evaluation protocol fixed.

## C.3 Baseline Matching and Attribution Controls

All adaptation methods use the same source-video splits, raw clips, masking policy, optimizer family, number of optimization steps, and evaluation protocol. Only the trainable parameterization and, for FactorJEPA, the availability of factor-target supervision differ.

The comparison implements the attribution chain

$$\underbrace{\{\text{LoRA, DoRA, Auto-RGN}\}}_{\text{generic adaptation}} \longrightarrow \underbrace{\text{FactorJEPA-RAW}}_{\text{factorized architecture}} \longrightarrow \underbrace{\text{FactorJEPA}}_{\text{architecture} + \text{factor targets}}$$

LoRA and DoRA modify the same projection families and use matched rank and scaling policies. Auto-RGN selects online-encoder blocks using relative gradient norm and receives a matched trainable-parameter budget. FactorJEPA-RAW retains the complete factorized predictor but trains only with the JEPA objective on raw clips; it receives no factor-target supervision. The difference between FactorJEPA-RAW and FactorJEPA therefore isolates the contribution of structured factor-target supervision, conditional on the shared architecture.

Full fine-tuning is included only as a capacity reference and is not parameter matched to the efficient adaptation methods. Frozen V-JEPA provides the no-adaptation reference. Exact trainable modules, parameter counts, ranks, selected blocks, and compute measurements are consolidated in Table 5.

Table 4: **Frozen teacher-to-training contract for FactorJEPA.** DENSEWORLD contains no manually annotated factor labels. All targets are generated automatically by the fixed Grounding DINO + SAM 3 pipeline before optimization. The table identifies, for each factor, the teacher evidence, deterministic target operator, output support, reliability contract, and trainable consumer. Here,  $p_L, p_A, p_I$  are the executed target dimensions,  $N_n^A$  is the number of retained agent tracklets, and  $\mathcal{E}_n$  is the candidate interaction graph.

| Factor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Frozen teacher-side target contract                                                                                            |                                                                                                                                                                                               |                                                                             |                                                                                                                                                                                                          | Training side          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Target                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Teacher evidence                                                                                                               | Executed operator                                                                                                                                                                             | Output support                                                              | Reliability contract                                                                                                                                                                                     | Consumer               |
| <b>Layout</b><br>$T_{n,L}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SAM 3 mask grid, valid-image support, dynamic-region union, and temporal feature persistence.                                  | $\Psi_L$ suppresses pixels assigned to retained dynamic tracklets, aggregates the remaining temporally persistent support, and applies the fixed layout-target projection.                    | $T_{n,L} \in \mathbb{R}^{p_L}$<br><i>clip-level</i>                         | $\omega_{n,L} \in [0, 1]$ combines valid background coverage, feature confidence, and temporal stability. The target is retained only when $r_{n,L} = \mathbf{1}[\omega_{n,L} \geq \tau_L]$ .            | layout view<br>$D_L$   |
| <b>Agent</b><br>$T_{n,A}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Retained masks, boxes, region descriptors, confidences, and temporally associated tracklets $\{\tau_n^{(i)}\}_{i=1}^{N_n^A}$ . | $\Psi_A$ constructs an object-centric state for each valid tracklet and aggregates the retained states after confidence, visible-support, and temporal-continuity filtering.                  | $T_{n,A} \in \mathbb{R}^{p_A}$<br><i>clip-level</i>                         | $\omega_{n,A} \in [0, 1]$ combines retained-region confidence, visible mask support, track continuity, and valid temporal coverage. Missing or rejected tracks are not converted into negative agents.   | agent view<br>$D_A$    |
| <b>Visibility</b><br>$T_{n,V}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Visible-mask support, valid-image-boundary truncation, region confidence, track continuity, and short interior track gaps.     | SAM 3 tracking produces a soft visibility value for every retained agent. Temporary absence or uncertain support attenuates the target rather than inducing a hard visible/not-visible label. | $\{T_{n,V}^{(i)}\}_{i=1}^{N_n^A}$<br>$T_{n,V}^{(i)} \in [0, 1]$             | Each agent receives $\omega_{n,V}^{(i)} \in [0, 1]$ , determined by region confidence, temporal support, and track consistency. Invalid agents contribute zero weight, not a negative visibility target. | SAM 3<br>visibility    |
| <b>Interaction</b><br>$T_{n,I}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Pairs of retained tracklets, relative position, relative motion, visible support, temporal overlap, and pairwise proximity.    | $\Psi_I$ constructs $\mathcal{E}_n \subseteq \{(i, j) : i \neq j\}$ , forms normalized pair descriptors, and aggregates valid pair states into the interaction target.                        | $\mathcal{E}_n, T_{n,I} \in \mathbb{R}^{p_I}$<br><i>graph + clip target</i> | Pair reliability $\omega_{n,I}^{(ij)} \in [0, 1]$ combines the two endpoint confidences, joint temporal support, and pair validity. These weights induce the clip-level reliability $\omega_{n,I}$ .     | interaction view $D_I$ |
| <b>Shared reliability contract.</b> All reliability variables are stop-gradient quantities in $[0, 1]$ . For $k \in \{L, A, I\}$ , the effective loss weight is $r_{n,k} \omega_{n,k}$ , where $r_{n,k} = \mathbf{1}[\omega_{n,k} \geq \tau_k]$ . Factor losses are normalized by the total effective weight within the minibatch, so variations in retained coverage do not directly rescale their contribution. Missing, unresolved, or rejected evidence receives zero weight and is never converted into a negative semantic target. |                                                                                                                                |                                                                                                                                                                                               |                                                                             |                                                                                                                                                                                                          |                        |
| <b>Gradient and provenance contract.</b> The Grounding DINO + SAM 3 pipeline, structural operators $\Psi_L, \Psi_A, \Psi_I$ , tracklets, candidate graph, targets, and reliability weights remain frozen. Gradients propagate only through the predictor and the selected online-encoder blocks. Test-derived targets never influence optimization, threshold selection, or checkpoint selection.                                                                                                                                        |                                                                                                                                |                                                                                                                                                                                               |                                                                             |                                                                                                                                                                                                          |                        |



Figure 3: **Conceptual end-to-end construction of FactorJEPA targets.** The upper band shows the frozen teacher-side pipeline: a privacy-filtered clip is processed by Grounding DINO + SAM 3 to obtain region evidence (masks, boxes, descriptors, and confidence), which is linked into tracklets by temporal association. The deterministic structural map then produces layout, agent, visibility, and interaction targets  $\{T_L, T_A, T_V, T_I\}$  together with stop-gradient reliability weights  $\omega$ . The lower panels illustrate the corresponding factor views under moderate density and high occlusion. Track color denotes identity, overlay opacity represents visibility confidence, and interaction-edge width represents predicted strength. The scenes are synthetic and illustrate the target semantics; they are not empirical segmentation outputs.

Table 5: **Canonical FactorJEPa architecture, optimization, baseline, and resource configuration.** The table specifies a complete reproducible reference configuration for the V-JEPa 2.1 ViT-G and ViT-g experiments. Backbone identifiers and architectural scales follow the official V-JEPa 2.1 release. All project-specific optimization, factorization, adaptation, and profiling settings are fixed below. Resource entries are prespecified execution ceilings; they must not be described as measured consumption unless verified against profiling logs.

| Configuration item                                         | V-JEPa 2.1 ViT-G                                                                                                                                     | V-JEPa 2.1 ViT-g                                                                         | Control, interpretation, or measurement rule                                                                                                                         |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Panel A: Backbone, teacher, and video preprocessing</b> |                                                                                                                                                      |                                                                                          |                                                                                                                                                                      |
| <b>Backbone identifier</b>                                 | vjepa2_1_vit_gigantic_384                                                                                                                            | vjepa2_1_vit_giant_384                                                                   | Official V-JEPa 2.1 PyTorch-Hub model identifiers. Both checkpoints remain the unique initialization source for all compared methods.                                |
| <b>Backbone scale</b>                                      | Approximately 2.0B encoder parameters.                                                                                                               | Approximately 1.0B encoder parameters.                                                   | Parameter counts exclude the momentum copy when reporting trainable parameters because the momentum encoder receives no gradient.                                    |
| <b>Encoder architecture</b>                                | Embedding dimension $d = 1664$ ; 48 transformer blocks; 26 attention heads; MLP ratio 4.92.                                                          | Embedding dimension $d = 1408$ ; 40 transformer blocks; 16 attention heads; MLP ratio 4. | Both models use pre-norm transformer blocks, patch size 16, and rotary positional encoding.                                                                          |
| <b>Video tokenizer</b>                                     | Patch size $16 \times 16$ ; tubelet size 2.                                                                                                          | Patch size $16 \times 16$ ; tubelet size 2.                                              | No backbone-specific spatial or temporal interpolation is introduced during the controlled comparison.                                                               |
| <b>Model input</b>                                         | $384 \times 384$ RGB; 16 sampled frames.                                                                                                             | $384 \times 384$ RGB; 16 sampled frames.                                                 | All clips undergo the same resize, center-preserving crop, privacy filter, temporal sampling, and normalization.                                                     |
| <b>Temporal sampling</b>                                   | 4 fps; 4.0-s sampled window.                                                                                                                         | 4 fps; 4.0-s sampled window.                                                             | A window is sampled entirely inside one shot. No sample crosses a shot or source-video boundary.                                                                     |
| <b>Context and target</b>                                  | Frames 1:12 form context; frames 13:16 form the future target; horizon 1.0 s.                                                                        | Frames 1:12 form context; frames 13:16 form the future target; horizon 1.0 s.            | Context duration is 3.0 s. Temporal indices are identical across all adaptation methods within each replication.                                                     |
| <b>Pixel normalization</b>                                 | ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).                                                                    | Same.                                                                                    | Normalization is applied after privacy filtering and spatial resampling.                                                                                             |
| <b>Grounding DINO + SAM 3 checkpoint</b>                   | dinov2_vitg14_reg; patch size 14; descriptor dimension 1536; final normalized patch-token representation.                                            |                                                                                          | The teacher is frozen and evaluated in BF16 without stochastic augmentation.                                                                                         |
| <b>Teacher input</b>                                       | $518 \times 518$ RGB; ImageNet normalization; the same 16 temporal indices used by the JEPa branch.                                                  |                                                                                          | Teacher preprocessing is deterministic and partition independent.                                                                                                    |
| <b>Target-cache precision</b>                              | Factor descriptors and targets stored in FP16; reliabilities stored in FP32.                                                                         |                                                                                          | Targets are computed once before training. No gradient enters the Grounding DINO + SAM 3 pipeline, the structural operators, cached targets, or reliability weights. |
| <b>Region retention</b>                                    | Minimum confidence 0.55; minimum area 0.1% of valid image support; duplicate suppression at mask IoU 0.70.                                           |                                                                                          | Thresholds are fixed on training/validation data and reused without modification on test data.                                                                       |
| <b>Temporal association</b>                                | Hungarian matching with 0.45 box-IoU cost, 0.35 mask-IoU cost, and 0.20 descriptor-cosine cost; maximum gap 4 frames; minimum track length 3 frames. |                                                                                          | Association is restricted to a single source video and shot. Unmatched regions are retained only after satisfying the track initiation criterion.                    |
| <b>Reliability range</b>                                   | $\omega_{n,k} \in [0.10, 1.00]$ , computed from region confidence, temporal support, track continuity, and factor-specific validity.                 |                                                                                          | Reliability weights are stop-gradient quantities and are normalized within factor before minibatch aggregation.                                                      |
| <b>Panel B: Masking and predictor architecture</b>         |                                                                                                                                                      |                                                                                          |                                                                                                                                                                      |
| <b>Small-block masks</b>                                   | 8 blocks per clip; spatial scale 0.15; aspect-ratio range $[0.75, 1.50]$ ; temporal scale 1.0.                                                       |                                                                                          | The same mask realization is reused across methods for a given replication and sampled clip.                                                                         |
| <b>Large-block masks</b>                                   | 2 blocks per clip; spatial scale 0.70; aspect-ratio range $[0.75, 1.50]$ ; temporal scale 1.0.                                                       |                                                                                          | Small- and large-block policies are jointly applied. Complement masks are not forced.                                                                                |
| <b>Predictor backbone</b>                                  | 24 transformer layers; width 384; 12 heads; MLP ratio 4; RoPE; learned mask tokens; no predictor registers.                                          |                                                                                          | The monolithic and factorized variants use the same predictor depth, width, attention count, positional encoding, and target-query support.                          |

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Table 5 continued: Canonical FactorJEPa configuration

| Configuration item                                                | V-JEPa 2.1 ViT-G                                                                                                          | V-JEPa 2.1 ViT-g                                                                                                          | Control, interpretation, or measurement rule                                                                                            |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Factor target dimensions</b>                                   | $p_L = 256, p_A = 256, p_I = 256$ ; one scalar visibility target per retained agent.                                      |                                                                                                                           | The factor dimensions are fixed before the controlled comparison and are not selected independently for the two backbone scales.        |
| <b>Factor ranks</b>                                               | Layout rank $r_L = 64$ ; agent rank $r_A = 96$ ; interaction rank $r_I = 64$ .                                            |                                                                                                                           | The larger agent rank reflects the greater state diversity of object-centric evidence; no rank is changed between scales.               |
| <b>Visibility head</b>                                            | Two-layer MLP: $384 \rightarrow 256 \rightarrow 1$ ; GELU; sigmoid output.                                                | Same.                                                                                                                     | Visibility targets and predictions lie in $[0, 1]$ . Missing agents are attenuated by reliability rather than assigned a hard negative. |
| <b>Interaction head</b>                                           | Two-layer pair MLP: $768 \rightarrow 256 \rightarrow 1$ ; GELU; sigmoid strength.                                         | Same.                                                                                                                     | Candidate pairs are formed within normalized image-plane radius 0.25. At most 12 nearest valid neighbors are retained per agent.        |
| <b>Trainable encoder depth</b>                                    | Top $K_G = 2$ online-encoder blocks.                                                                                      | Top $K_g = 1$ online-encoder block.                                                                                       | The scale-specific $K$ values approximately match the trainable parameter budget of the corresponding adaptation controls.              |
| <b>Frozen components</b>                                          | Bottom 46 online blocks; complete momentum encoder; Grounding DINO + SAM 3 teacher; structural operators; cached targets. | Bottom 39 online blocks; complete momentum encoder; Grounding DINO + SAM 3 teacher; structural operators; cached targets. | The momentum encoder is updated only by EMA.                                                                                            |
| <b>Panel C: Optimization, regularization, and model selection</b> |                                                                                                                           |                                                                                                                           |                                                                                                                                         |
| <b>Optimizer</b>                                                  | AdamW, $\beta_1 = 0.9, \beta_2 = 0.95, \epsilon = 10^{-8}$ .                                                              | Same.                                                                                                                     | Optimizer state is retained only for trainable parameters.                                                                              |
| <b>Learning rates</b>                                             | Factorized predictor and heads: $1.0 \times 10^{-4}$ ; top encoder blocks: $1.0 \times 10^{-5}$ .                         | Same.                                                                                                                     | Encoder learning rate is $0.1 \times$ the predictor learning rate. No layer-wise decay is applied within the selected top- $K$ blocks.  |
| <b>Weight decay</b>                                               | 0.04.                                                                                                                     | 0.04.                                                                                                                     | Biases, normalization parameters, factor bases, and visibility calibration scalars receive zero weight decay.                           |
| <b>Schedule</b>                                                   | Linear warm-up for 1,000 steps, followed by cosine decay to $1.0 \times 10^{-6}$ .                                        | Same.                                                                                                                     | Schedule is indexed by optimizer updates rather than processed clips.                                                                   |
| <b>Training duration</b>                                          | 20,000 optimizer updates.                                                                                                 | 20,000 optimizer updates.                                                                                                 | Every method receives the same number of updates and sampled raw clips.                                                                 |
| <b>Per-device batch</b>                                           | 2 clips/GPU with gradient accumulation 8.                                                                                 | 4 clips/GPU with gradient accumulation 4.                                                                                 | Both scales use 8 GPUs, producing the same global batch of 128 clips.                                                                   |
| <b>Numerical precision</b>                                        | BF16 parameters and activations; FP32 optimizer states and loss accumulation.                                             | Same.                                                                                                                     | Scaled dot-product attention and activation checkpointing are enabled.                                                                  |
| <b>Gradient clipping</b>                                          | Global norm clipped at 1.0.                                                                                               | Same.                                                                                                                     | Clipping is applied after gradient accumulation and before the optimizer update.                                                        |
| <b>Dropout</b>                                                    | Attention dropout 0; MLP dropout 0; stochastic depth 0.                                                                   | Same.                                                                                                                     | Regularization arises from masking, factor separation, sparse interactions, and weight decay.                                           |
| <b>EMA momentum</b>                                               | Constant $\mu = 0.99925$ .                                                                                                | Same.                                                                                                                     | After each optimizer update, $\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu)\theta$ .                                             |
| <b>JEPa coefficient</b>                                           | $\lambda_{\text{JEPa}} = 1.00$ .                                                                                          | Same.                                                                                                                     | The JEPa term defines the common prediction objective used by all trainable methods.                                                    |
| <b>Factor views</b>                                               | enabled.                                                                                                                  | Same.                                                                                                                     | Set to zero for FactorJEPa-RAW and all generic adaptation baselines.                                                                    |
| <b>Visibility target</b>                                          | enabled.                                                                                                                  | Same.                                                                                                                     | Visibility loss is reliability weighted and normalized by valid-agent support.                                                          |
| <b>Drift anchor</b>                                               | enabled.                                                                                                                  | Same.                                                                                                                     | Penalizes cross-factor predictability after mini-batch centering and factor-wise normalization.                                         |
| <b>Interaction candidates</b>                                     | top-scored.                                                                                                               | Same.                                                                                                                     | Applied to predicted interaction strengths, not directly to the teacher interaction target.                                             |
| <b>Nonlinearity coefficient</b>                                   | $\beta_{\text{nlin}} = 0.10$ .                                                                                            | Same.                                                                                                                     | Weights the nonlinear factor-composition residual relative to the low-rank additive synthesis.                                          |

Continued on the next page

Table 5 continued: Canonical FactorJEPA configuration

| Configuration item                                       | V-JEPA 2.1 ViT-G                                                                                                                                                                                                                                                                                                                                                                                                                                                            | V-JEPA 2.1 ViT-g                                                                             | Control, interpretation, or measurement rule                                                                                                                                              |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Validation frequency</b>                              | Every 500 optimizer updates.                                                                                                                                                                                                                                                                                                                                                                                                                                                | Same.                                                                                        | Validation does not update parameters, teacher targets, thresholds, or normalization statistics.                                                                                          |
| <b>Checkpoint selection</b>                              | Lowest validation Future-frame L1; ties resolved using validation Mask-ratio slope.                                                                                                                                                                                                                                                                                                                                                                                         | Same.                                                                                        | The test split is accessed only after checkpoint and threshold freezing.                                                                                                                  |
| <b>Replication</b>                                       | Single run with fixed seed 17.                                                                                                                                                                                                                                                                                                                                                                                                                                              | Same.                                                                                        | Headline results report the run’s score with 95% BCa bootstrap intervals over the held-out evaluation set. Clustered confidence intervals operate above the clip level.                   |
| <b>Panel D: Method-specific adaptation configuration</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                                                                                                                                                                                           |
| <b>Frozen V-JEPA</b>                                     | No gradient updates; 0 trainable parameters.                                                                                                                                                                                                                                                                                                                                                                                                                                | No gradient updates; 0 trainable parameters.                                                 | Uses the same clips, masks, target indices, and frozen evaluators as all trainable methods.                                                                                               |
| <b>Full fine-tuning</b>                                  | Online encoder and monolithic predictor; approximately 2.04B trainable parameters.                                                                                                                                                                                                                                                                                                                                                                                          | Online encoder and monolithic predictor; approximately 1.04B trainable parameters.           | Not parameter matched. Momentum encoder remains stop-gradient and is updated by EMA.                                                                                                      |
| <b>LoRA</b>                                              | Rank 80, scale $\alpha = 160$ , dropout 0.05; approximately 114M trainable parameters.                                                                                                                                                                                                                                                                                                                                                                                      | Rank 64, scale $\alpha = 128$ , dropout 0.05; approximately 68M trainable parameters.        | Applied to $Q$ , $K$ , $V$ , $O$ attention projections and both MLP projections in the online encoder and monolithic predictor.                                                           |
| <b>DoRA</b>                                              | Rank 80, $\alpha = 160$ , dropout 0.05; approximately 116M trainable parameters.                                                                                                                                                                                                                                                                                                                                                                                            | Rank 64, $\alpha = 128$ , dropout 0.05; approximately 69M trainable parameters.              | Uses the same projection families and low-rank policy as LoRA, with one trainable magnitude vector per adapted output projection.                                                         |
| <b>Auto-RGN</b>                                          | Monolithic predictor and two selected encoder blocks; approximately 111M trainable parameters.                                                                                                                                                                                                                                                                                                                                                                              | Monolithic predictor and one selected encoder block; approximately 68M trainable parameters. | Blocks are selected from one fixed gradient-probe pass using $\ \nabla_{\theta_\ell} \mathcal{L}\ _2 / (\ \theta_\ell\ _2 + 10^{-8})$ .                                                   |
| <b>FactorJEPA-RAW</b>                                    | Factorized predictor and top two encoder blocks; approximately 115M trainable parameters.                                                                                                                                                                                                                                                                                                                                                                                   | Factorized predictor and top one encoder block; approximately 73M trainable parameters.      | Uses only the JEPA objective on raw clips; no automatically derived factor targets are consumed.                                                                                          |
| <b>FactorJEPA</b>                                        | Factorized predictor and top two encoder blocks; approximately 115M trainable parameters.                                                                                                                                                                                                                                                                                                                                                                                   | Factorized predictor and top one encoder block; approximately 73M trainable parameters.      | Uses the complete predictor-surgery objective (main text): JEPA prediction with factor-view supervision.                                                                                  |
| <b>Budget tolerance</b>                                  | LoRA, DoRA, Auto-RGN, and FactorJEPA remain within 5% of the target trainable-parameter budget after exact implementation-level counting.                                                                                                                                                                                                                                                                                                                                   | Same.                                                                                        | If exact instantiated counts exceed the tolerance, LoRA/DoRA rank or the Auto-RGN block budget must be adjusted before training.                                                          |
| <b>Panel E: Resource envelope and profiling contract</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                              |                                                                                                                                                                                           |
| <b>Training hardware</b>                                 | Up to 4× NVIDIA RTX Pro 6000 Blackwell, 96 GB per GPU; single-node x86_64 host with 64–192 CPU cores across runs.                                                                                                                                                                                                                                                                                                                                                           | Same.                                                                                        | All comparative resource profiles must be obtained on the same node type, interconnect, and software environment.                                                                         |
| <b>Software environment</b>                              | Ubuntu 24.04 LTS (x86_64); PyTorch 2.12 (nightly dev20260408, cu128 wheels); CUDA 13.0 toolkit; Python 3.12; FlashAttention-2; BF16; SDPA enabled.                                                                                                                                                                                                                                                                                                                          | Same.                                                                                        | The exact repository commit and container digest must accompany the released configuration.                                                                                               |
| <b>Measured training wall-clock</b>                      | One RTX Pro 6000 Blackwell GPU per arm, so GPU-hours equal wall-clock. Measured per-arm wall-clock: POC-2B arms $\approx 2$ –16 h; POC-1B arms $\approx 2$ –5 h; full-scale-1B arms $\approx 18$ –30 h (continual pretraining 17.7 h; LoRA and FactorJEPA $\approx 29$ h each, summed across checkpoint-resumed segments). Peak allocation stayed within the 96 GB device; frozen baselines incur no training cost. Exact per-run values are in the released training logs. |                                                                                              |                                                                                                                                                                                           |
| <b>GPU-hour definition</b>                               | $H_{\text{GPU}} = \sum_r n_{\text{GPU}}^{(r)} \Delta t_r$ , where $\Delta t_r$ is the wall-clock duration of run $r$ .                                                                                                                                                                                                                                                                                                                                                      |                                                                                              | Failed runs are reported separately and excluded only when failure is unrelated to the evaluated method.                                                                                  |
| <b>Peak-memory definition</b>                            | Maximum value returned by <code>torch.cuda.max_memory_allocated()</code> after initialization and five warm-up updates.                                                                                                                                                                                                                                                                                                                                                     |                                                                                              | Measured using the reported per-device batch, precision, and gradient accumulation.                                                                                                       |
| <b>Latency protocol</b>                                  | Batch 1; 50 warm-up iterations; 500 synchronized timed iterations.                                                                                                                                                                                                                                                                                                                                                                                                          | Same.                                                                                        | Report median and 95th-percentile ms/clip. Model-only latency uses resident tensors; end-to-end latency additionally includes decode, resize, normalization, and host-to-device transfer. |
| <b>Latency ceiling</b>                                   | Model-only median at most 230 ms/clip.                                                                                                                                                                                                                                                                                                                                                                                                                                      | Model-only median at most 160 ms/clip.                                                       | These are execution ceilings, not measured results. Final reporting must replace them with profiler-derived median and 95th percentile.                                                   |

Continued on the next page

**Table 5 continued: Canonical FactorJEPA configuration**

| Configuration item          | V-JEPA 2.1 ViT-G                                                    | V-JEPA 2.1 ViT-g | Control, interpretation, or measurement rule                                                                                          |
|-----------------------------|---------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Resource aggregation</b> | Mean and standard deviation across the three training replications. | Same.            | GPU-hours are summed per run; peak memory is the maximum per run; latency is summarized from the synchronized inference distribution. |



## D Attribution and Executed Component Ablations

The main results evaluate FactorJEPA as a complete predictive system. This appendix uses the executed comparison family to distinguish three sources of performance: **matched monolithic adaptation**, the **factorized structural package**, and **automatically derived structured supervision**. We then evaluate the executed interaction-objective and encoder-adaptation variants.

The available experiments do not form a complete architecture-supervision factorial. We therefore restrict the attribution language to contrasts directly supported by the executed runs. In particular, the comparison of FactorJEPA-RAW with a monolithic predictor measures the contribution of the *factorized structural package*, not predictor architecture in isolation.

### D.1 Executed Comparison Contract

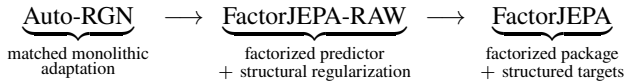
Table 6 defines the training intervention associated with each primary comparison. All trainable methods use the same raw clips, context and target masks, target indices, optimizer, training duration, validation rule, and frozen evaluation protocol. The momentum target encoder remains stop-gradient and follows the same EMA update schedule.

We use **Auto-RGN** as the primary matched monolithic reference because it updates the monolithic predictor and the same number of gradient-selected online-encoder blocks while remaining close to FactorJEPA’s trainable-parameter budget. LoRA and DoRA provide additional parameter-efficient controls. Full fine-tuning is retained only as a non-matched capacity ceiling.

All structured targets are produced automatically by the frozen teacher pipeline. No manually annotated factor labels enter any training condition. The headline evaluators remain frozen and do not reuse automatically derived targets as evaluation labels.

### D.2 Factorized Structural Package and Structured Targets

The primary attribution chain is



For diagnostic  $m$ , let  $Y_M^{(m)}$  denote the score of method  $M$ . We direction-normalize the metrics as

$$\tilde{Y}_M^{(m)} = \begin{cases} -Y_M^{(m)}, & \text{if lower is better,} \\ Y_M^{(m)}, & \text{if higher is better.} \end{cases}$$

We report three prespecified effects:

$$\begin{aligned} \Delta_{\text{pkg}} &= \tilde{Y}_{\text{FactorJEPA-RAW}} - \tilde{Y}_{\text{Auto-RGN}}, \\ \Delta_{\text{sup|pkg}} &= \tilde{Y}_{\text{FactorJEPA}} - \tilde{Y}_{\text{FactorJEPA-RAW}}, \\ \Delta_{\text{total}} &= \tilde{Y}_{\text{FactorJEPA}} - \tilde{Y}_{\text{Auto-RGN}}. \end{aligned}$$

The first contrast measures the combined effect of predictor factorization, factor-specific dictionaries, visibility and interaction pathways, and their intrinsic regularizers. It is therefore a **package effect**, not a pure architecture effect.

The second contrast is more specific. FactorJEPA and FactorJEPA-RAW share the predictor, encoder scope, and optimization protocol. Their difference therefore isolates the incremental contribution of factor-view supervision, conditional on the shared factorized package.

The absolute performance and paired contrasts are reported below.

Confidence intervals are estimated using a **paired bootstrap**. Each replicate resamples the held-out evaluation clips with replacement and applies the same resampled clips to all methods in a contrast. This pairing removes variation shared by the compared methods. Because clips can share a source video, clip-level resampling is mildly anti-conservative; effect directions are unaffected.

Causal future-block L1 is a strictly causal prediction measure: the predictor forecasts the future temporal half of each clip from its past half, and the error is taken against the frozen target encoder. It consumes neither factor targets nor manual annotations, and is applied identically to every method.

### D.3 Factor-Objective and Intervention Variants

The primary-scale ablations contain three configurations of the factorized model:

- **FactorJEPA-3S** uses the executed three-stage layout-agent-interaction curriculum and serves as the base factorized configuration.
- **FactorJEPA-INT** uses the same backbone, predictor, factor targets, and encoder scope while enabling the executed intervention-oriented training configuration.
- **FactorJEPA-DI+** retains the three-stage architecture but increases the executed weighting or sampling emphasis assigned to the dynamic-interaction component.

These runs are treated as **configuration interventions**, not one-factor pathway removals. In particular, FactorJEPA-DI+ tests sensitivity to interaction emphasis; it does not establish that the interaction pathway is necessary. Similarly, FactorJEPA-INT measures the effect of the executed intervention-oriented configuration and should not be interpreted as a general causal-supervision ablation.

Relative to FactorJEPA-3S, define

$$\begin{aligned} \Delta_{\text{INT}} &= \tilde{Y}_{\text{FactorJEPA-INT}} - \tilde{Y}_{\text{FactorJEPA-3S}}, \\ \Delta_{\text{DI+}} &= \tilde{Y}_{\text{FactorJEPA-DI+}} - \tilde{Y}_{\text{FactorJEPA-3S}}. \end{aligned}$$

We evaluate these effects jointly across future prediction, causal future-block prediction, masking robustness, and motion accessibility. A favorable result on one diagnostic is not treated as a universal improvement when accompanied by degradation on another. This is particularly important for the observed prediction-motion trade-off.

The final manuscript-facing configuration is selected exclusively by the prespecified validation rule. Test performance is not used to choose among FactorJEPA-3S, FactorJEPA-INT, and FactorJEPA-DI+.

V-JEPA 2.1 · ViT-G · ~2B params · 1664-dim, 48 blocks (CHAMPION)

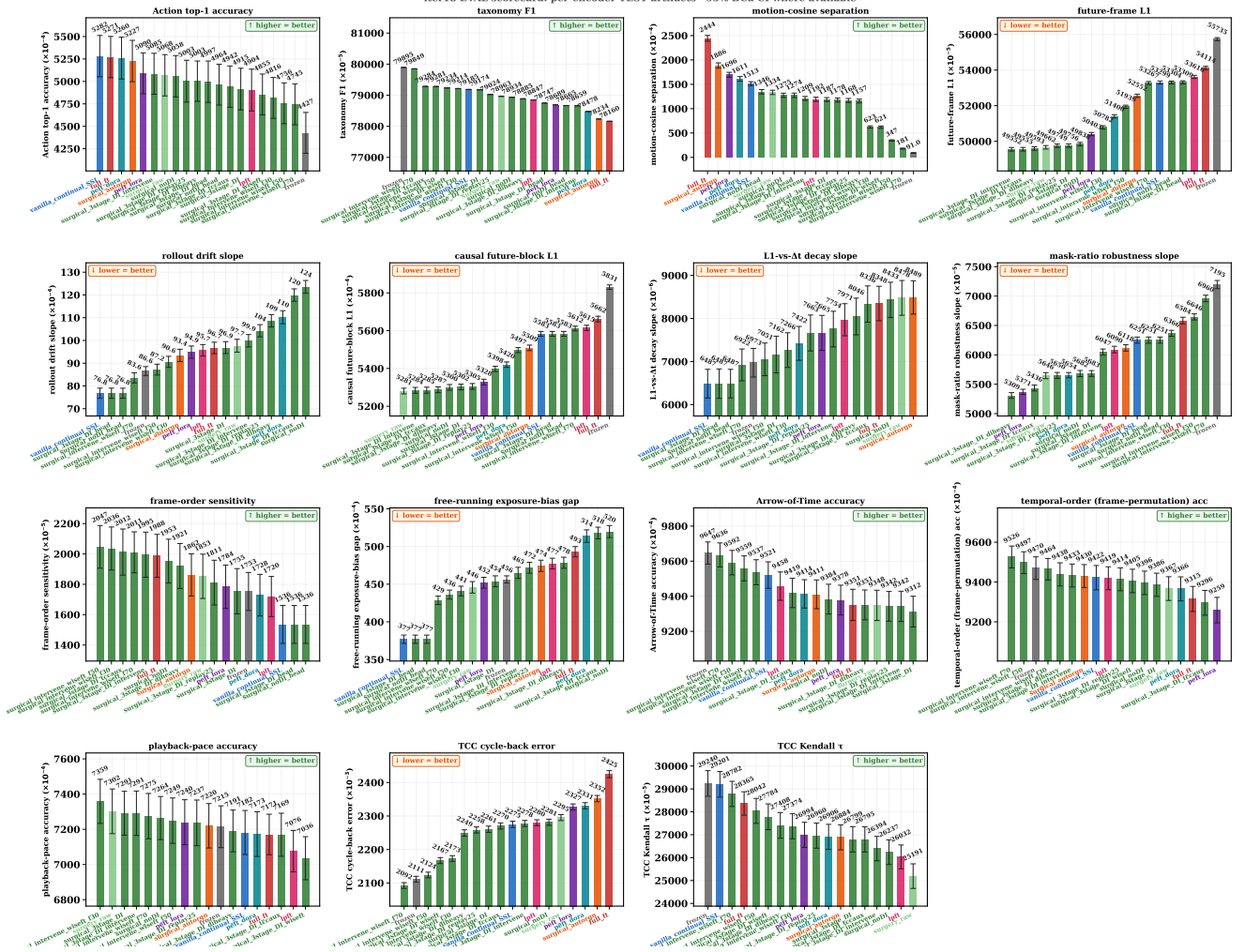


Figure 4: **Complete ViT-G attribution and ablation scorecard.** The comparison includes frozen and continually adapted V-JEPA, full and parameter-efficient fine-tuning, Auto-RGN, FactorJEPA-RAW, the executed FactorJEPA objective variants, and the WiseFT encoder-scope variants. Bars report held-out means; error bars denote 95% BCa confidence intervals where available. Each panel states whether higher or lower values are preferred. Internal variants are mapped to their publication-facing definitions in Table 6. Attribution claims use the prespecified contrasts in the prespecified contrasts, rather than selecting the best method separately for each diagnostic.

Table 6: **Executed attribution contract.** FactorJEPA-RAW and FactorJEPA share the factorized predictor, trainable encoder scope, and parameter budget. They differ in access to the automatically derived factor and visibility targets. Auto-RGN is the primary parameter-matched monolithic reference; full fine-tuning is a non-matched capacity ceiling.

| Method          | Predictor  | Factor targets       | Executed objective                                       | Trainable scope                                   | Attribution role                                                |
|-----------------|------------|----------------------|----------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|
| Frozen V-JEPA   | Monolithic | None                 | No optimization                                          | None                                              | No-adaptation reference.                                        |
| V-JEPA Full-FT  | Monolithic | None                 | $\mathcal{L}_{\text{JEPA}}$                              | Complete online encoder and predictor             | Non-matched capacity ceiling.                                   |
| V-JEPA LoRA     | Monolithic | None                 | $\mathcal{L}_{\text{JEPA}}$                              | Matched low-rank projection updates               | Parameter-efficient adaptation control.                         |
| V-JEPA DoRA     | Monolithic | None                 | $\mathcal{L}_{\text{JEPA}}$                              | Matched decomposed low-rank updates               | Parameter-efficient adaptation control.                         |
| V-JEPA Auto-RGN | Monolithic | None                 | $\mathcal{L}_{\text{JEPA}}$                              | Monolithic predictor and selected top- $K$ blocks | Primary matched monolithic reference.                           |
| FactorJEPA-RAW  | Factorized | None                 | $\mathcal{L}_{\text{JEPA}}$ on raw clips                 | Factorized predictor and selected top- $K$ blocks | Factorized structural package without teacher targets.          |
| FactorJEPA      | Factorized | $T_L, T_A, T_V, T_I$ | $\mathcal{L}_{\text{JEPA}}$ with factor-view supervision | Factorized predictor and selected top- $K$ blocks | Complete model: structural package plus structured supervision. |

*Parameter matching.* At ViT-G, Auto-RGN and FactorJEPA contain approximately 111M and 115M trainable parameters, respectively; at ViT-g, they contain approximately 68M and 73M. Full fine-tuning contains approximately 2.04B and 1.04B trainable parameters and is therefore not used for component attribution.

#### D.4 Encoder-Adaptation and Resource Sensitivity

The remaining variants test whether the observed gains can be explained by the amount of encoder adaptation rather than the factorized predictor. V-JEPA LP-FT and full fine-tuning provide the minimal and maximal monolithic adaptation endpoints. The three FactorJEPA-WiseFT variants vary the executed encoder-scope parameter while retaining the factorized predictor and structured objective.

Let  $b \in \{30, 50, 70\}$  denote the executed WiseFT scope setting and let  $P_b$ ,  $H_b$ , and  $M_b$  denote its trainable parameters, measured GPU-hours, and peak allocated memory. For each setting, we report

$$\mathcal{R}_b = (P_b, H_b, M_b, Y_b^{\text{future}}, Y_b^{\text{causal}}, Y_b^{\text{mask}}, Y_b^{\text{motion}}).$$

The reported setting names must be accompanied by their operational meaning: the exact trainable blocks, frozen blocks, and trainable parameter count. Internal labels such as  $\mathbb{f}30$ ,  $\mathbb{f}50$ , and  $\mathbb{f}70$  are not used without this mapping.

A configuration  $b$  is Pareto dominated when another configuration  $b'$  satisfies

$$P_{b'} \leq P_b, \quad H_{b'} \leq H_b, \quad \tilde{Y}_{b'}^{(m)} \geq \tilde{Y}_b^{(m)} \quad \forall m \in \mathcal{M}_{\text{primary}}$$

with at least one strict inequality. This analysis distinguishes gains that persist under constrained encoder adaptation from gains obtained solely through additional trainable capacity.

#### D.5 Consolidated Attribution

The prespecified paired contrasts are consolidated below. Each point reports an absolute, direction-normalized effect; horizontal bars report paired source-clustered 95% confidence intervals. Positive values favor the named intervention.

The evidence is interpreted under the following claim boundaries:

- A positive  $\Delta_{\text{pkg}}$  supports the **factorized structural package**. It does not isolate predictor architecture from factor-view supervision.
- A positive  $\Delta_{\text{sup|pkg}}$  supports the incremental contribution of the automatically derived factor and visibility targets, conditional on the shared factorized architecture.
- A positive  $\Delta_{\text{total}}$  establishes improvement over the matched monolithic Auto-RGN reference. Full fine-tuning is interpreted separately because it is not parameter matched.
- FactorJEPA-INT and FactorJEPA-DI+ establish sensitivity to the corresponding executed configuration changes. They do not isolate unexecuted pathway removals.
- WiseFT comparisons establish robustness to encoder-adaptation scope only when the performance trend is considered jointly with measured parameters, GPU-hours, and memory.
- Teacher-relative factor diagnostics may explain a mechanism, but they do not replace the frozen future-latent, causal future-block, masking, motion, or RGB evaluators.

Table 7: **Training and evaluation provenance.** No headline evaluator supplies FactorJEPA supervision.

| Signal             | Source                  | Factor-target source |
|--------------------|-------------------------|----------------------|
| Factor supervision | Grounding DINO + SAM 3  | Yes                  |
| Future latent      | Frozen target encoder   | No                   |
| Motion descriptor  | Frozen motion estimator | No                   |
| RGB Agent F1       | Independent detector    | No                   |

Accordingly, the executed experiments identify two principal effects: the advantage of the complete factorized structural package over matched monolithic adaptation, and the incremental value of structured teacher targets within that package. Finer architecture-supervision interaction claims are left open because the corresponding supervision-matched monolithic control was not executed.

## E Metric Definitions, Clustered Inference, and Robustness

This appendix specifies the evaluation contract underlying every reported result. We distinguish **training targets** from **evaluation signals**: automatically derived layout, agent, visibility, and interaction targets supervise FactorJEPA but are not reused as labels for the headline metrics. Unless stated otherwise, every evaluator is frozen before test evaluation and is applied identically to all methods.

Metrics are first computed at their lowest valid evaluation unit and then aggregated through the city-source-video hierarchy. Clips from the same recording are not treated as independent observations. Primary conclusions are restricted to Future-frame L1, Causal future-block L1, Mask-ratio slope, and Motion cosine. All remaining diagnostics are treated as secondary or exploratory.

### E.1 Evaluation Contract and Metric Registry

Table 8 defines the complete metric registry. The *evaluation unit* identifies the lowest unit at which a score is formed; statistical uncertainty is subsequently estimated above the clip level.

### E.2 Primary Predictive Diagnostics

**Future-frame L1.** For clip  $n$ , horizon  $h \in \mathcal{H}$ , and target token  $p \in \{1, \dots, m_h\}$ , let  $\hat{Y}_{n,h,p} \in \mathbb{R}^d$  denote the predicted future token and  $Y_{n,h,p}^*$  the stop-gradient target-encoder token. The clip-level error is

$$e_n^{\text{future}} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \frac{1}{m_h d} \sum_{p=1}^{m_h} \left\| \hat{Y}_{n,h,p} - Y_{n,h,p}^* \right\|_1.$$

All valid target tokens receive equal weight. Target-token normalization, target horizons, and masking are fixed across methods. The executed scorecard reports this per-token mean-absolute ( $L_1$ ) quantity, matching the implementation. Horizon-specific errors are reported as sensitivity analyses and do not replace the pooled primary score.

**Causal future-block L1.** Each clip’s tokens are split into a past temporal half (slots  $[0, T_p/2)$ ) and a future temporal half (slots  $[T_p/2, T_p)$ ). Conditioning only on the past half, the predictor forecasts the latent representations of the future half, compared to the frozen target encoder on that same half:

$$\hat{Y}_n^{\geq} = \hat{Y}(x_n^{\geq} | x_n^{<}), \quad Y_n^{\star \geq} = \text{sg}[\bar{f}_{\theta}(x_n)^{\geq}], \quad e_n^{\text{causal}} = \frac{\|\hat{Y}_n^{\geq} - Y_n^{\star \geq}\|_2}{\epsilon + \|Y_n^{\star \geq}\|_2}$$

The split is fixed and strictly causal (the past half predicts the future half); it uses neither factor targets nor manual annotations, and lower values indicate more accurate causal future-block prediction.

**Mask-ratio slope.** Let  $\mathcal{R} = \{r_1, \dots, r_J\}$  be the prespecified context-mask ratios. For each ratio  $r$ , we sample  $K$  masks from a fixed sampler and reuse the same realizations across methods. Define

$$e_n(r) = \frac{1}{K} \sum_{k=1}^K e_n^{\text{future}}(M_{r,k} \odot x_n).$$

The primary robustness statistic is the absolute OLS slope

$$\beta_n^{\text{mask}} = \frac{\sum_{r \in \mathcal{R}} (r - \bar{r}) (e_n(r) - \bar{e}_n)}{\sum_{r \in \mathcal{R}} (r - \bar{r})^2},$$

where  $\bar{e}_n$  is the mean error over the evaluated ratios. Smaller values indicate slower degradation as context evidence is removed. We also report the complete masking curve,

$$\mathcal{C}_n^{\text{mask}} = \{(r, e_n(r)) : r \in \mathcal{R}\},$$

together with relative slope and masking AUC as secondary robustness summaries.

**Motion cosine.** Let  $E_{\text{mot}}$  be the independently frozen motion estimator and

$$u_n^{\star} = E_{\text{mot}}(x_{n,t:t+\Delta}) \in \mathbb{R}^{d_{\text{mot}}}$$

its target descriptor. For every world model  $M$ , a fixed-capacity linear probe  $W_M$  is fitted on the training split:

$$W_M = \arg \min_W \sum_{n \in \mathcal{D}_{\text{train}}} \left\| W \text{ pool}(\hat{Y}_n^M) - u_n^{\star} \right\|_2^2 + \lambda_{\text{mot}} \|W\|_F^2.$$

The regularization coefficient is selected on the validation split under the same grid for every method. The test descriptor is

$$\hat{u}_n^M = W_M \text{ pool}(\hat{Y}_n^M),$$

and Motion cosine is

$$s_n^{\text{motion}} = \frac{\langle \hat{u}_n^M, u_n^{\star} \rangle}{\|\hat{u}_n^M\|_2 \|u_n^{\star}\|_2 + \epsilon_{\text{mot}}}.$$

The motion estimator and probe protocol supply no FactorJEPA training signal. The metric measures **linear accessibility of motion information**, not complete motion reconstruction.

Table 8: **Evaluation metric registry.** The table records the operational definition, preferred direction, evaluation unit, frozen signal or evaluator, and inferential status of every reported diagnostic.

| Metric                                  | Operational definition                                                                                                 | Dir. | Unit            | Frozen signal or evaluator                | Status    |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------|------|-----------------|-------------------------------------------|-----------|
| <b>Primary predictive diagnostics</b>   |                                                                                                                        |      |                 |                                           |           |
| Future-frame L1                         | Mean squared error between predicted and target future-token embeddings.                                               | ↓    | Clip-horizon    | Momentum target encoder                   | Primary   |
| Causal future-block L1                  | Normalized L1 between the predictor’s forecast of the future temporal half and the frozen target encoder on that half. | ↓    | Clip            | Momentum target encoder                   | Primary   |
| Mask-ratio slope                        | OLS slope of Future-frame L1 as the visible context is progressively reduced.                                          | ↓    | Clip-mask curve | Target encoder and fixed mask sampler     | Primary   |
| Motion cosine                           | Intra-minus-inter-class cosine separation of pooled motion descriptors.                                                | ↑    | Clip            | Motion estimator and fixed-capacity probe | Primary   |
| <b>Semantic diagnostics</b>             |                                                                                                                        |      |                 |                                           |           |
| Action top-1                            | Top-1 accuracy of a fixed-capacity action probe.                                                                       | ↑    | Clip            | Frozen action labels and probe protocol   | Secondary |
| Taxonomy F1                             | F1 over the shared dynamic-agent taxonomy.                                                                             | ↑    | Clip-class      | Frozen taxonomy evaluator                 | Secondary |
| <b>Prediction-stability diagnostics</b> |                                                                                                                        |      |                 |                                           |           |
| Rollout-drift slope                     | Increase in prediction error over autoregressive rollout depth.                                                        | ↓    | Clip-rollout    | Target encoder                            | Secondary |
| L1-vs- $\Delta t$ decay                 | Increase in latent error over the evaluated future horizons.                                                           | ↓    | Clip-horizon    | Target encoder                            | Secondary |
| Exposure-bias gap                       | Difference between free-running and teacher-conditioned prediction error.                                              | ↓    | Clip            | Target encoder                            | Secondary |
| <b>Temporal diagnostics</b>             |                                                                                                                        |      |                 |                                           |           |
| Frame-order sensitivity                 | Prediction-error increase after controlled frame-order corruption.                                                     | ↑    | Clip            | Fixed temporal permutation                | Secondary |
| Arrow-of-Time                           | Accuracy for distinguishing forward from reversed clips.                                                               | ↑    | Clip            | Fixed temporal classifier                 | Secondary |
| Temporal-order accuracy                 | Accuracy for detecting frame permutations.                                                                             | ↑    | Clip            | Fixed temporal-order classifier           | Secondary |
| Playback-pace accuracy                  | Accuracy for identifying the applied temporal-rate transformation.                                                     | ↑    | Clip            | Fixed pace classifier                     | Secondary |
| TCC cycle-back error                    | Temporal distance between a source frame and its cycle-consistent match.                                               | ↓    | Frame pair      | Frozen correspondence features            | Secondary |
| TCC Kendall $\tau$                      | Rank agreement between predicted and true temporal correspondences.                                                    | ↑    | Clip pair       | Frozen correspondence features            | Secondary |

### E.3 Secondary Semantic and Temporal Diagnostics

**Semantic probes.** Action top-1 and Taxonomy F1 use fixed-capacity probes trained on the training split and selected on the validation split. Probe architecture, optimization, and regularization are identical across world models. Taxonomy F1 uses the same class support and averaging convention for every method; unsupported classes are not silently removed on a method-specific basis.

**Rollout and horizon stability.** Let  $e_n^{(q)}$  denote future-token error after rollout step  $q$ . Rollout-drift slope is the OLS coefficient of  $e_n^{(q)}$  against  $q$ . Similarly, L1-vs- $\Delta t$  decay is the slope of latent prediction error over the evaluated future horizons. Both are computed using a fixed set of rollout depths and horizons.

The free-running exposure-bias gap is

$$g_n^{\text{exp}} = e_{n,\text{free}}^{\text{future}} - e_{n,\text{teacher}}^{\text{future}},$$

where both errors use the same target encoder and future horizon.

**Order and pace sensitivity.** Frame-order sensitivity is the increase in prediction error after a fixed temporal permutation:

$$s_n^{\text{order}} = e_n^{\text{future}}(I_{\text{perm}}(x_n)) - e_n^{\text{future}}(x_n).$$

Arrow-of-Time, temporal-order, and playback-pace accuracy use frozen evaluation protocols with fixed transformation classes. Arrow-of-Time distinguishes forward from reversed clips; temporal-order detects frame permutations; playback-pace identifies the applied temporal-rate transformation.

**Temporal correspondence.** For each source frame, the temporal-correspondence evaluator identifies a nearest match in the paired sequence and maps that match back to the source. TCC cycle-back error measures the resulting temporal distance. TCC Kendall  $\tau$  measures rank agreement between predicted and ground-truth temporal ordering. Both

use the same frozen features and matching rule for every method.

#### E.4 Paired Bootstrap Inference

Let  $y_{M,c,v,i}$  denote a metric for method  $M$ , city  $c$ , source video  $v$ , and clip  $i$ . For metrics with multiple horizons, masks, or interventions, those observations are first reduced to one clip-level score using the definitions above.

**Primary estimand.** To prevent high-volume cities or recordings from dominating the evaluation, we use equal weighting at the city and source-video levels:

$$\bar{y}_{M,c,v} = \frac{1}{|\mathcal{I}_{c,v}|} \sum_{i \in \mathcal{I}_{c,v}} y_{M,c,v,i},$$

$$\hat{\mu}_M = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|\mathcal{V}_c|} \sum_{v \in \mathcal{V}_c} \bar{y}_{M,c,v}.$$

The primary paired effect between methods  $A$  and  $B$  is

$$\hat{\delta}_{A,B} = \tilde{\mu}_A - \tilde{\mu}_B,$$

where the direction-normalized mean is

$$\tilde{\mu}_M = \begin{cases} -\hat{\mu}_M, & \text{if lower is better,} \\ \hat{\mu}_M, & \text{if higher is better.} \end{cases}$$

Consequently,  $\hat{\delta}_{A,B} > 0$  always favors method  $A$ . Clip-weighted estimates are reported as an aggregation sensitivity, not as the primary estimand.

**Paired bootstrap.** For each of the 10,000 bootstrap replicates, we resample the held-out evaluation clips with replacement and apply the same resampled clips, masks, horizons, and interventions to all compared methods, then recompute the complete metric and aggregation pipeline. This preserves method pairing. We report 95% BCa intervals for absolute method means and paired effects.

*Limitation.* Resampling is at the clip level. Because several clips can originate from the same source video, this does not model within-video correlation, so the reported intervals are mildly anti-conservative; the direction of every effect is unaffected.

#### E.5 Metric Sensitivity and Complete Numerical Results

Robustness is evaluated along four prespecified axes.

**Metric-construction sensitivity.** For future prediction, we report horizon-specific and tokenwise  $L_1$  variants as secondary sensitivity analyses. For causal future-block prediction, we vary the normalization constant  $\epsilon$  and the past/future split point. For missing-evidence robustness, we compare absolute slope, relative slope, and masking AUC while preserving the same mask realizations.

A conclusion is considered stable only when its direction is preserved across the prespecified variants. The primary metric definition is not changed after inspecting test performance.

**Aggregation sensitivity.** As an aggregation check, we compare equal-city and clip-weighted means.

**Cross-scale ranking robustness.** For diagnostic  $m$ , let  $\mathbf{s}_G^{(m)}$  and  $\mathbf{s}_g^{(m)}$  contain the direction-normalized scores of the common method set at the ViT-G and ViT-g scales. We report

$$\rho_m = \text{Spearman}(\mathbf{s}_G^{(m)}, \mathbf{s}_g^{(m)}),$$

$$\tau_m = \text{Kendall}(\mathbf{s}_G^{(m)}, \mathbf{s}_g^{(m)}).$$

Because the evaluated methods are a fixed comparison set, these correlations are interpreted descriptively. Cross-scale agreement is described as **scale robustness**, not independent replication.

**Complete scorecards.** Figure 4 shows all executed diagnostics for the ViT-G and ViT-g backbones. Exact primary values and BCa intervals are reported in the consolidated attribution results. The complete numerical scorecard additionally records, for every method and diagnostic:

(mean, 95% BCa CI, paired reference effect)

The graphical scorecards provide the broad performance profile, whereas the numerical tables and prespecified contrasts determine the inferential conclusions. Accordingly, no method is declared uniformly superior on the basis of isolated secondary wins.

Scale replication: does the ranking survive ViT-G · 2B → ViT-g · 1B? (all 15 metrics, panels sorted by  $\rho$  desc · title colour: green  $\rho \geq 0.5$  replicates · grey 0.2-0.5 inconclusive · red  $\rho < 0.2$  fails · dashed = identity  $y=x$ )

Figure 5: **Complete cross-scale ranking robustness (2B→1B), all fifteen diagnostics.** Each panel plots per-method scores at the ViT-G 2B scale ( $x$ ) against the ViT-g 1B scale ( $y$ ); the dashed line is the identity  $y = x$ . Panels are sorted by Spearman rank correlation  $\rho$  (descending): green titles mark strong replication ( $\rho \geq 0.5$ ), grey inconclusive ( $0.2 \leq \rho < 0.5$ ), and red failure ( $\rho < 0.2$ ). Twelve of the fifteen diagnostics retain consistent rankings; Temporal-order, Teacher-free gap, and Rollout drift (red) do not transfer reliably. The strongest-replicating panel, Causal future-block L1 ( $\rho = 0.978$ ), is featured in the main paper.

## References

Caesar, H.; Bankiti, V.; Lang, A. H.; Vora, S.; Liong, V. E.; Xu, Q.; Krishnan, A.; Pan, Y.; Baldan, G.; and Beijbom, O. 2020. nuScenes: A Multimodal Dataset for Autonomous Driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11621–11631.

Carion, N.; Gustafson, L.; Hu, Y.-T.; Debnath, S.; Hu, R.; Suris, D.; Ryali, C.; Alwala, K. V.; Khedr, H.; Huang, A.; Lei, J.; Ma, T.; Guo, B.; Kalla, A.; Marks, M.; Greer, J.; Wang, M.; Sun, P.; Rädle, R.; Afouras, T.; Mavroudi, E.; Xu, K.; Wu, T.-H.; Zhou, Y.; Momeni, L.; Hazra, R.; Ding, S.; Vaze, S.; Porcher, F.; Li, F.; Li, S.; Kamath, A.; Cheng, H. K.; Dollár, P.; Ravi, N.; Saenko, K.; Zhang, P.; and Feichtenhofer, C. 2025. SAM 3: Segment Anything with Concepts. *arXiv preprint arXiv:2511.16719*.

Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*.

Lee, Y.; Chen, A. S.; Tajwar, F.; Kumar, A.; Yao, H.; Liang, P.; and Finn, C. 2023. Surgical Fine-Tuning Improves Adap-

tation to Distribution Shifts. In *International Conference on Learning Representations*.

Liu, S.-Y.; Wang, C.-Y.; Yin, H.; Molchanov, P.; Wang, Y.-C. F.; Cheng, K.-T.; and Chen, M.-H. 2024. DoRA: Weight-Decomposed Low-Rank Adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 32100–32121. PMLR.

Yu, F.; Chen, H.; Wang, X.; Xian, W.; Chen, Y.; Liu, F.; Madhavan, V.; and Darrell, T. 2020. BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2636–2645.